On Schrödinger's Equation

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Abstract. We study Schrödinger's equation as a linear dispersive partial differential equation whose unknown is a complex wave function. We derive the equation from the correspondence between classical Hamiltonian mechanics and wave operators, explain why the wave function is normalized in L^2, prove the conservation of probability through the continuity equation, and solve the free-particle equation by Fourier transform. We then develop separation of variables, stationary states, spectral expansions, the infinite square well, and the harmonic oscillator from an applied mathematics perspective. The last part discusses finite-difference, Crank-Nicolson, spectral, split-step Fourier, and finite-element methods, emphasizing which discrete structures preserve mass, stability, and the oscillatory character of quantum evolution [Sch26, Bor26, Dir30, Sto32, CH53, RS72, RS75, Kat95, Eva10, Tes14, Tre00, BS08].

1 Introduction

Schrödinger's equation is one of the cleanest places where physics, Fourier analysis, spectral theory, and numerical partial differential equations meet. At the physical level, it describes the evolution of a quantum state. At the analytic level, it is a linear PDE with a skew-Hermitian time generator. At the numerical level, it is a demanding test case because the solution is complex-valued, oscillatory, dispersive, and norm-preserving [CH53, RS72, RS75, Eva10, Tes14].

We shall treat the equation as an applied mathematics object. This means that we will not begin with the full axiomatic structure of quantum mechanics, although that structure explains why the equation has the form it has [Dir30, vN32]. Instead, we begin with the classical energy relation, turn energy and momentum into differential operators, obtain the PDE, and then ask how one solves it. This route is historically close to Schrödinger's wave-mechanical viewpoint [Sch26], but we shall phrase the result in the modern language of Hilbert spaces and self-adjoint operators.

The guiding theme is that each useful solution method exposes a different diagonal structure. Plane waves diagonalize the free Hamiltonian, eigenfunctions diagonalize confined systems, Hermite functions diagonalize the quadratic potential, and unitary groups diagonalize the abstract evolution through the spectral theorem [Sto32, RS72, RS75, Kat95]. Numerically, the same principle reappears in finite-dimensional form. A stable method tries to preserve the Hermitian structure of the Hamiltonian rather than treating the equation as an arbitrary complex-valued evolution problem [CN47, Str68, Tre00].

*Version 0.1. This is a working note; corrections, sharper proofs, and missing references are welcome.

We proceed linearly. To begin with, we derive the PDE and isolate the assumptions that make the derivation meaningful. We then prove probability conservation, as the probabilistic interpretation of the wave function is useful only if the evolution preserves normalization. After that, we solve the free equation by Fourier transform, pass to stationary states and spectral expansions, compute the two standard solvable models, and finally explain how the same structural constraints guide numerical methods.

2 From Hamiltonian Mechanics to the PDE

We begin with the simplest non-relativistic particle of mass $m > 0$ moving in a scalar potential $V(x)$. The classical Hamiltonian is

$$E = \frac{p^2}{2m} + V(x), \tag{2.1}$$

where p is momentum and E is energy. Quantum mechanics replaces the state of the particle by a wave function

$$\psi : \mathbb{R}^d \times \mathbb{R} \to \mathbb{C}, \tag{2.2}$$

where $x \in \mathbb{R}^d$ is position and $t \in \mathbb{R}$ is time.

The important change is the passage from phase-space coordinates to an operator equation. Classically, position and momentum are coordinates on phase space, and the Hamiltonian is a scalar function on that phase space. In the wave formulation, momentum is represented by differentiation, the Hamiltonian becomes an operator, and the equation asks for a function whose time derivative is controlled by that operator. At this stage, the physical model has become a PDE problem.

2.1 Plane Waves and Operator Replacement

The shortest derivation uses a plane wave as a calibration device. Consider

$$\psi(x,t) = A \exp\left(\frac{\mathrm{i}}{\hbar}(p \cdot x - Et)\right), \tag{2.3}$$

where $\hbar$ is the reduced Planck constant. Differentiating Equation (2.3) gives

$$\mathrm{i}\hbar \frac{\partial \psi}{\partial t} = E\psi, \tag{2.4}$$

$$-\mathrm{i}\hbar \nabla \psi = p\psi. \tag{2.5}$$

Thus, the plane wave is an eigenfunction of the time derivative and of the spatial gradient. If we want the energy relation Equation (2.1) to act on general superpositions of such waves, then the formal replacements suggested by Equations (2.4) and (2.5) are

$$E \rightsquigarrow \mathrm{i}\hbar \frac{\partial}{\partial t}, \qquad p \rightsquigarrow -\mathrm{i}\hbar\nabla. \tag{2.6}$$

Definition 2.1 (Schrödinger Hamiltonian)**.** For a real-valued potential $V : \mathbb{R}^d \to \mathbb{R}$, the formal Schrödinger Hamiltonian is

$$H = -\frac{\hbar^2}{2m}\Delta + V(x), \tag{2.7}$$

where $\Delta = \sum_{j=1}^{d} \partial^2/\partial x_j^2$ is the Laplacian. ⌟

Applying Equation (2.6) to the classical Hamiltonian relation yields the time-dependent Schrödinger equation

$$\mathrm{i}\hbar \frac{\partial \psi}{\partial t}(x,t) = -\frac{\hbar^2}{2m}\Delta\psi(x,t) + V(x)\psi(x,t). \tag{2.8}$$

Equivalently, if H denotes Equation (2.7), then Equation (2.8) is

$$\mathrm{i}\hbar \frac{\partial \psi}{\partial t} = H\psi. \tag{2.9}$$

This compact notation hides two analytic choices. The first is the domain of H, since the Laplacian is not defined on every L^2-function. The second is the boundary condition, since the same differential expression can define different operators on an interval, a box, or all of space. These choices determine whether H is self-adjoint, and self-adjointness is the condition that later gives unitary time evolution [RS72, RS75, Kat95, Tes14].

2.2 What the Derivation Does and Does Not Prove

The preceding derivation is not a proof of quantum mechanics. It is a consistency argument showing that Equation (2.8) is the linear PDE whose plane-wave solutions satisfy the de Broglie and Planck relations. The real mathematical content begins after the equation is written down. We must specify the function space in which ψ lives, the domain on which H acts, and the boundary conditions imposed by the physical system.

For most of this note the natural ambient space is $L^2(\mathbb{R}^d)$, with inner product [RS72, Eva10, Tes14]

$$\langle f, g\rangle_{L^2} = \int_{\mathbb{R}^d} \overline{f(x)} g(x)\,\mathrm{d}x. \tag{2.10}$$

The normalization condition is

$$\int_{\mathbb{R}^d} |\psi(x,t)|^2\,\mathrm{d}x = 1. \tag{2.11}$$

In the probabilistic interpretation introduced by Born [Bor26], the quantity $|\psi(x,t)|^2$ is the probability density for the position of the particle at time t.

Therefore, the wave function itself is not a probability density, and its phase cannot be discarded. The density uses only $|\psi|^2$, but interference, dispersion, and stationary phases all depend on the complex phase of ψ. Much of the applied mathematics of Schrödinger's equation is the problem of tracking this phase accurately enough that the derived probability density remains meaningful [Dir30, Bor26, RS75, Tes14].

3 Probability Conservation

The first structural test for Equation (2.8) is whether it preserves Equation (2.11). The answer is yes, provided the Hamiltonian is real in the appropriate sense and boundary terms vanish. This conservation law is the PDE form of unitary time evolution.

We prove this first because it explains why Schr̀odinger's equation is not another heat equation. The heat equation dissipates mass in high frequencies and drives solutions toward smoother profiles. Schŕodinger's equation transports and interferes probability amplitude without dissipating the total L^2-mass. The distinction comes from the factor i, which turns a formally self-adjoint spatial operator into a skew-adjoint time generator [Eva10, RS72, Tes14].

Proposition 3.1 (Continuity equation)**.** Let ψ be a smooth solution of Equation (2.8) on $\mathbb{R}^d$, and assume that ψ and $\nabla\psi$ decay sufficiently fast at spatial infinity. Define

$$\rho(x,t) = |\psi(x,t)|^2, \qquad J(x,t) = \frac{\hbar}{m}\,\mathrm{Im}\left(\overline{\psi(x,t)}\nabla\psi(x,t)\right). \tag{3.1}$$

Then

$$\frac{\partial\rho}{\partial t} + \nabla\cdot J = 0. \tag{3.2}$$

In particular, the L^2-mass of ψ is conserved. ⌟

Proof. The proof uses only the equation and its complex conjugate. The potential term disappears because the potential is real-valued, and the remaining expression is a divergence. Hence, the local conservation law is a bookkeeping identity for how probability density flows through space.

We compute

$$\frac{\partial}{\partial t}|\psi|^2 = \overline{\psi}\frac{\partial\psi}{\partial t} + \psi\frac{\partial\overline{\psi}}{\partial t}. \tag{3.3}$$

From Equation (2.8) we obtain

$$\frac{\partial\psi}{\partial t} = \frac{\mathrm{i}\hbar}{2m}\Delta\psi - \frac{\mathrm{i}}{\hbar}V\psi. \tag{3.4}$$

Taking the complex conjugate of Equation (3.4), and using that V is real-valued, gives

$$\frac{\partial\overline{\psi}}{\partial t} = -\frac{\mathrm{i}\hbar}{2m}\Delta\overline{\psi} + \frac{\mathrm{i}}{\hbar}V\overline{\psi}. \tag{3.5}$$

Substituting Equations (3.4) and (3.5) into Equation (3.3), the potential terms cancel and we get

$$\frac{\partial}{\partial t}|\psi|^2 = \frac{\mathrm{i}\hbar}{2m}\overline{\psi}\Delta\psi - \frac{\mathrm{i}\hbar}{2m}\psi\Delta\overline{\psi} \tag{3.6}$$

$$= -\nabla\cdot\left(\frac{\hbar}{m}\,\mathrm{Im}\left(\overline{\psi}\nabla\psi\right)\right). \tag{3.7}$$

The last equality follows from the identity

$$\nabla\cdot\left(\overline{\psi}\nabla\psi - \psi\nabla\overline{\psi}\right) = \overline{\psi}\Delta\psi - \psi\Delta\overline{\psi}. \tag{3.8}$$

This proves Equation (3.2). Integrating Equation (3.2) over $\mathbb{R}^d$, using the divergence theorem, and using the assumed decay at infinity gives

$$\frac{}{\mathrm{d}t}\int_{\mathbb{R}^d}|\psi(x,t)|^2\,\mathrm{d}x = -\int_{\mathbb{R}^d}\nabla\cdot J(x,t)\,\mathrm{d}x = 0. \tag{3.9}$$

Thus, the normalization in Equation (2.11) is preserved. □

The continuity equation is stronger than global conservation. It identifies a local flux J, so that probability can leave a region only by crossing its boundary. On a bounded domain, this observation is where boundary conditions enter again. Dirichlet, Neumann, periodic, and absorbing boundary conditions do not have the same probability balance, and a numerical method that ignores this

$$\psi_0(x) \xrightarrow{\mathcal{F}} \widehat{\psi}_0(\xi) \longrightarrow e^{-\mathrm{i}\hbar t|\xi|^2/(2m)}\widehat{\psi}_0(\xi) \xrightarrow{\mathcal{F}^{-1}} \psi(x,t)$$

FIGURE 1. Fourier solution of the free Schrödinger equation. The PDE diagonalizes after the spatial Fourier transform, each Fourier coefficient solves a scalar ODE, and the inverse transform reconstructs the wave function.

distinction can produce a qualitatively wrong evolution even when its pointwise truncation error is small [RS75, Tes14, BS08].

4 The Free Particle and Fourier Solution

The free equation is the case $V = 0$. It is the quantum analogue of a particle with no external force, but analytically it is more revealing to view it as a dispersive PDE. Unlike the heat equation, it does not smooth by damping high frequencies. Instead, each Fourier mode rotates with a frequency proportional to $|\xi|^2$ [CH53, Eva10, RS75].

This is the first place where the wave nature of the equation becomes visible. Fourier modes do not decay, and so the solution operator is unitary on L^2. Nevertheless, a localized packet spreads because different frequencies rotate at different angular velocities. Dispersion is therefore not loss of mass. It is a loss of coherence among phases [RS75, Eva10, Tes14].

$$\mathrm{i}\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\Delta\psi, \qquad \psi(x,0) = \psi_0(x). \tag{4.1}$$

Let $\widehat{\psi}(\xi,t) = \mathcal{F}\psi(\xi,t)$. Applying the Fourier transform in x to Equation (4.1) gives

$$\mathrm{i}\hbar\frac{\partial\widehat{\psi}}{\partial t}(\xi,t) = \frac{\hbar^2\,|\xi|^2}{2m}\widehat{\psi}(\xi,t). \tag{4.2}$$

For each fixed ξ, Equation (4.2) is an ordinary differential equation. Solving it yields

$$\widehat{\psi}(\xi,t) = \exp\left(-\mathrm{i}\frac{\hbar t}{2m}\,|\xi|^2\right)\widehat{\psi}_0(\xi). \tag{4.3}$$

Thus, the free solution is

$$\psi(x,t) = \mathcal{F}^{-1}\left[\exp\left(-\mathrm{i}\frac{\hbar t}{2m}\,|\xi|^2\right)\widehat{\psi}_0(\xi)\right](x). \tag{4.4}$$

Figure 1 explains why the Fourier transform is the natural method for the free equation. It turns spatial differentiation into multiplication, and therefore turns the PDE into independent scalar evolutions indexed by frequency.

One should also note what this method does not use. It does not use compactness, boundary conditions, or a discrete list of eigenvalues. The free Hamiltonian has continuous spectrum, and the Fourier transform is the device that replaces a sum over eigenvectors by an integral over frequencies [RS75, Kat95, Tes14].

For $t \neq 0$, the inverse transform gives the integral kernel

$$\psi(x,t) = \left(\frac{m}{2\pi \mathrm{i}\hbar t}\right)^{d/2} \int_{\mathbb{R}^d} \exp\left(\frac{\mathrm{i}m\,|x-y|^2}{2\hbar t}\right) \psi_0(y)\,\mathrm{d}y, \tag{4.5}$$

with the branch chosen continuously from $t > 0$. Formula Equation (4.5) shows dispersion directly. The amplitude has size proportional to $|t|^{-d/2}$, while the phase oscillates quadratically in $|x-y|$.

The kernel formula is often the most useful applied expression. It says that the value at x and time t is assembled from all starting points y, with rapidly oscillating phases selecting the dominant contributions. This is the elementary ancestor of stationary phase approximations and semiclassical asymptotics [CH53, Eva10, Tes14].

5 Stationary States and Separation of Variables

Fourier methods solve translation-invariant problems. When the potential or the domain destroys translation invariance, the next method is separation of variables [CH53, Eva10, Tes14]. We look for solutions of the form

$$\psi(x,t) = \phi(x)T(t). \tag{5.1}$$

Substituting Equation (5.1) into Equation (2.9) gives

$$\mathrm{i}\hbar\phi(x)T'(t) = T(t)H\phi(x). \tag{5.2}$$

Dividing by $\phi(x)T(t)$, where the quotient is defined, gives

$$\mathrm{i}\hbar\frac{T'(t)}{T(t)} = \frac{H\phi(x)}{\phi(x)}. \tag{5.3}$$

The left-hand side depends only on t, and the right-hand side depends only on x. Hence, both sides must be equal to a constant E. We obtain the stationary eigenvalue problem

$$H\phi = E\phi \tag{5.4}$$

and the scalar time equation

$$\mathrm{i}\hbar T'(t) = ET(t). \tag{5.5}$$

Solving Equation (5.5) gives

$$T(t) = \exp\left(-\mathrm{i}Et/\hbar\right). \tag{5.6}$$

Thus, every eigenfunction ϕ of H produces a solution

$$\psi(x,t) = \exp\left(-\mathrm{i}Et/\hbar\right)\phi(x). \tag{5.7}$$

The probability density of a stationary state is independent of time, since

$$|\psi(x,t)|^2 = |\phi(x)|^2. \tag{5.8}$$

This is why such states are called stationary. The wave function still rotates in the complex plane, but the observable position density does not change. Superpositions behave differently.

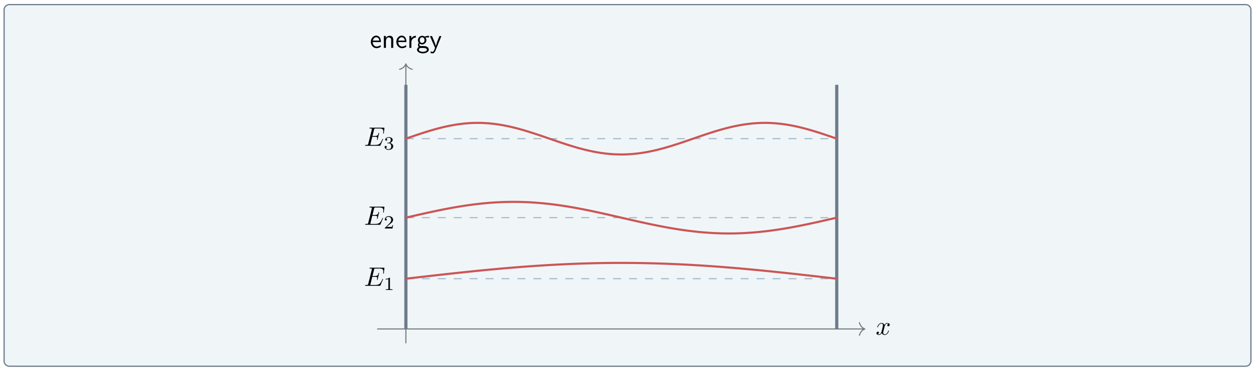

FIGURE 2. The infinite square well converts Schrödinger's equation into a Sturm-Liouville eigenvalue problem. The allowed energies are discrete because the boundary conditions permit only standing waves with an integer number of half-wavelengths.

If two eigenfunctions have different energies, their relative phase changes in time, and the resulting interference pattern can change even though each component separately has a stationary density [Dir30, RS75, Tes14].

The separation method is not limited to one eigenfunction. If $\{\phi_n\}_{n\geq 1}$ is an orthonormal basis of eigenfunctions with $H\phi_n = E_n\phi_n$, then an initial state

$$\psi_0 = \sum_{n\geq 1} c_n \phi_n \tag{5.9}$$

evolves as

$$\psi(x,t) = \sum_{n\geq 1} c_n \exp\left(-\mathrm{i}E_n t/\hbar\right) \phi_n(x). \tag{5.10}$$

Figure 2 is the canonical example of quantization by boundary conditions. Discreteness of energy is not inserted by hand. It arises because the spatial differential operator has a discrete spectrum on a bounded interval.

The hypothesis that the eigenfunctions form a basis is a spectral statement, not a formal consequence of separation of variables. In regular bounded one-dimensional problems it follows from Sturm-Liouville theory. In higher-dimensional or unbounded problems, the spectrum may have both discrete and continuous parts, and the expansion must be interpreted through the spectral theorem rather than through an ordinary Fourier series [CH53, RS72, RS75, Tes14].

6 The Infinite Square Well

Consider a particle constrained to the interval $(0, L)$, with infinite potential walls at $x = 0$ and $x = L$. Mathematically, this means that we solve the free equation on $(0, L)$ with Dirichlet boundary conditions [CH53, RS75, Tes14]

$$\phi(0) = 0, \qquad \phi(L) = 0. \tag{6.1}$$

The stationary problem is

$$-\frac{\hbar^2}{2m}\phi''(x) = E\phi(x), \qquad 0 < x < L. \tag{6.2}$$

Writing

$$k^2 = \frac{2mE}{\hbar^2}, \tag{6.3}$$

we obtain

$$\phi''(x) + k^2\phi(x) = 0. \tag{6.4}$$

The general solution of Equation (6.4) is

$$\phi(x) = A\sin(kx) + B\cos(kx). \tag{6.5}$$

The boundary condition $\phi(0) = 0$ forces $B = 0$. The boundary condition $\phi(L) = 0$ then gives

$$A\sin(kL) = 0. \tag{6.6}$$

For a nonzero eigenfunction we require $A \neq 0$, and hence $kL = n\pi$ for an integer $n \geq 1$. Therefore

$$E_n = \frac{\hbar^2\pi^2n^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots. \tag{6.7}$$

Normalizing in $L^2(0, L)$ gives

$$\phi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right). \tag{6.8}$$

Thus, the solution with initial state $\psi_0 \in L^2(0, L)$ is

$$\psi(x,t) = \sum_{n=1}^{\infty} c_n \exp\left(-\mathrm{i}E_n t/\hbar\right)\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), \tag{6.9}$$

where

$$c_n = \sqrt{\frac{2}{L}}\int_0^L \psi_0(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x. \tag{6.10}$$

The calculation shows how the wave function enters PDE solving. The initial wave function is decomposed into spatial eigenmodes, each eigenmode evolves by a phase factor, and the resulting phases interfere when the modes are summed again.

The infinite-well model should also be read as an idealization. Infinite walls encode a limiting boundary condition rather than a physically infinite force that one literally applies. Mathematically, this idealization is valuable because it isolates the role of boundary constraints, and it gives a complete solvable model in which quantization follows from the requirement that the wave vanish at both endpoints [RS75, Tes14].

7 The Harmonic Oscillator

The harmonic oscillator is the model potential [Dir30, RS75, Tes14]

$$V(x) = \frac{1}{2}m\omega^2x^2. \tag{7.1}$$

Its stationary equation on the line is

$$-\frac{\hbar^2}{2m}\phi''(x) + \frac{1}{2}m\omega^2 x^2\phi(x) = E\phi(x). \tag{7.2}$$

After the nondimensional change of variables

$$y = \sqrt{\frac{m\omega}{\hbar}}x, \tag{7.3}$$

Equation (7.2) becomes

$$-\phi''(y) + y^2\phi(y) = \lambda\phi(y), \qquad \lambda = \frac{2E}{\hbar\omega}. \tag{7.4}$$

The square-integrable solutions are Hermite functions, and the corresponding energies are [Dir30, RS75, Tes14]

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \dots. \tag{7.5}$$

The ground state is Gaussian,

$$\phi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right). \tag{7.6}$$

This example is important because it is both exactly solvable and structurally stable. Near a non-degenerate minimum of a smooth potential, the quadratic Taylor approximation gives a harmonic oscillator, so Equation (7.5) is the local model for small oscillations [CH53, RS75, Tes14].

There is a second reason the oscillator is central. The Hamiltonian factors into creation and annihilation operators after nondimensionalization, and this algebraic factorization explains the equally spaced energy levels. Thus, the oscillator teaches a different lesson from the square well. The square well quantizes through boundary conditions, whereas the oscillator quantizes through the confining growth of the potential and the algebra of the resulting differential operator [Dir30, RS75, Tes14].

8 Hilbert Space Formulation

The preceding sections solve concrete equations. The abstract theory explains why these solution formulas are not accidents. If H is self-adjoint on a dense domain in L^2, then Stone's theorem gives a unitary group [Sto32, RS72, Kat95]

$$U(t) = \exp\left(-\mathrm{i}tH/\hbar\right), \qquad t \in \mathbb{R}, \tag{8.1}$$

and the solution is

$$\psi(t) = U(t)\psi_0. \tag{8.2}$$

Self-adjointness is the analytic condition that makes the Hamiltonian a legitimate generator of quantum evolution [Sto32, RS72, Tes14].

It is useful to distinguish self-adjointness from the weaker statement that the differential expression looks symmetric after integration by parts. A symmetric operator may fail to be self-adjoint if its domain is too small or if boundary conditions have not been specified correctly. In quantum mechanics, this distinction affects the model itself. Different self-adjoint extensions can correspond to different physical boundary behavior, and only a self-adjoint Hamiltonian gives a uniquely defined unitary evolution for all times [vN32, RS72, RS75, Kat95, Tes14].

Theorem 8.1 (Unitary evolution, after Stone)**.** Let H be self-adjoint on a dense domain $D(H) \subseteq L^2(\mathbb{R}^d)$. Then there exists a strongly continuous unitary group $U(t)$ satisfying Equation (8.1). For every $\psi_0 \in D(H)$, the function $\psi(t) = U(t)\psi_0$ solves Equation (2.9) in L^2, and

$$\|\psi(t)\|_{L^2} = \|\psi_0\|_{L^2} \tag{8.3}$$

for every $t \in \mathbb{R}$. ⌟

We use Theorem 8.1 as a structural result rather than proving Stone's theorem here [Sto32, RS72, Kat95]. The point is that the same object controls physical interpretation and analytic well-posedness. The wave function remains normalized because the time-evolution operator is unitary.

This abstract formulation also unifies the explicit methods already discussed. Fourier solution, sine-series expansion, Hermite expansion, and matrix exponentiation are all instances of applying $e^{-\mathrm{i}tH/\hbar}$ after choosing a representation in which H is diagonal or nearly diagonal [RS75, Tes14, Tre00].

9 Numerical Solution Methods

Numerical methods for Schrödinger's equation should respect the features that make the continuous equation special. A good method does more than approximate derivatives. It should also control the L^2-norm, handle oscillatory phases, and treat the kinetic and potential parts in a way that matches the problem scale [CN47, Str68, Tre00, BS08].

The central numerical issue is that local consistency alone is not enough. A scheme can approximate the derivatives in Equation (2.8) and still drift in mass, introduce artificial damping, or generate phase errors that destroy interference patterns over long times. For this reason, structure-preserving schemes are often preferable to superficially simpler explicit schemes [CN47, Str68, Tre00].

9.1 Finite Differences

On a one-dimensional grid $x_j = jh$, the second derivative can be approximated by [CN47, Eva10]

$$\frac{\partial^2 \psi}{\partial x^2}(x_j, t) \approx \frac{\psi_{j+1}(t) - 2\psi_j(t) + \psi_{j-1}(t)}{h^2}. \tag{9.1}$$

This gives a semi-discrete system

$$\mathrm{i}\hbar \frac{\mathrm{d}\boldsymbol{\psi}}{\mathrm{d}t} = \mathbf{H}\boldsymbol{\psi}, \tag{9.2}$$

where $\mathbf{H}$ is a Hermitian matrix when the boundary conditions are imposed symmetrically. The exact semi-discrete flow is unitary because

$$\boldsymbol{\psi}(t) = \exp\left(-\mathrm{i}t\mathbf{H}/\hbar\right)\boldsymbol{\psi}(0). \tag{9.3}$$

Figure 3 shows the local nature of the finite-difference method. Each grid value interacts only with its nearest neighbors for the standard second-difference Laplacian.

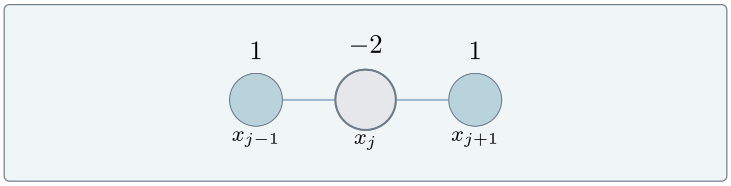

Figure 3. A finite-difference discretization replaces the spatial Laplacian by a nearest-neighbor stencil. The resulting ODE system is finite-dimensional and has a Hermitian Hamiltonian matrix when the discrete operator is assembled symmetrically.

The finite-difference discretization should be interpreted as a Galerkin-free way of producing a finite-dimensional Hamiltonian system. Once the spatial operator becomes a Hermitian matrix, the remaining question is how to integrate a finite-dimensional unitary ODE without spoiling the Hermitian structure [CN47, Tre00].

9.2 Crank-Nicolson

The Crank-Nicolson method applies the trapezoidal rule to Equation (9.2) [CN47]. With time step Δt, it reads

$$\left(\mathbf{I} + \frac{\mathrm{i}\Delta t}{2\hbar}\mathbf{H}\right)\boldsymbol{\psi}^{n+1} = \left(\mathbf{I} - \frac{\mathrm{i}\Delta t}{2\hbar}\mathbf{H}\right)\boldsymbol{\psi}^{n}. \tag{9.4}$$

Thus

$$\boldsymbol{\psi}^{n+1} = \left(\mathbf{I} + \frac{\mathrm{i}\Delta t}{2\hbar}\mathbf{H}\right)^{-1}\left(\mathbf{I} - \frac{\mathrm{i}\Delta t}{2\hbar}\mathbf{H}\right)\boldsymbol{\psi}^{n}. \tag{9.5}$$

When $\mathbf{H}$ is Hermitian, the matrix in Equation (9.5) is unitary. This is the main reason the method is preferred over explicit Euler for Schrödinger evolution.

The method can also be viewed as the Cayley transform of the Hermitian matrix $\mathbf{H}$. This interpretation explains why it preserves norm exactly at the semi-discrete level, while explicit Euler does not. The price is that each time step requires solving a linear system, but the gain is stability that matches the qualitative behavior of the continuous equation [CN47, Tre00].

9.3 Split-Step Fourier

When

$$H = H_{\mathrm{kin}} + H_{\mathrm{pot}}, \qquad H_{\mathrm{kin}} = -\frac{\hbar^2}{2m}\Delta, \qquad H_{\mathrm{pot}} = V(x), \tag{9.6}$$

the kinetic part is diagonal in Fourier space, and the potential part is diagonal in physical space. Strang splitting uses the approximation [Str68, Tre00]

$$e^{-\mathrm{i}\Delta t(H_{\mathrm{kin}}+H_{\mathrm{pot}})/\hbar} = e^{-\mathrm{i}\Delta t H_{\mathrm{pot}}/(2\hbar)}e^{-\mathrm{i}\Delta t H_{\mathrm{kin}}/\hbar}e^{-\mathrm{i}\Delta t H_{\mathrm{pot}}/(2\hbar)} + \mathcal{O}\left(\Delta t^3\right). \tag{9.7}$$

Accordingly, one time step multiplies by a potential phase, Fourier transforms, multiplies by a kinetic phase, inverse Fourier transforms, and multiplies by the potential phase again.

This method is efficient when the domain and boundary conditions allow fast Fourier transforms. It is also geometrically natural, since each substep is unitary and therefore the composition is unitary up to the splitting error.

The split-step method is most effective when the potential term is easy in physical space, and the kinetic term is easy in frequency space. It fails to be a universal method precisely because these two conveniences depend on geometry. On irregular domains, or with boundary conditions not compatible with the Fourier basis, finite elements or problem-adapted spectral bases are often more natural [Tre00, BS08].

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